Talk 1.

Thanks a lot for participating this ~~seminar~~. reading seminar.

Today I plan to motivate and sketsch a bit what we are going to do.

There are so many things that I want to mention a bit,

but today let's focus on the most basic questions.

- Why v.b. / l.b. ?
- What category? (better than Vect_X)
- Which examples ?

$X/\mathbb{C}$: sm proj var (in this talk).

Why: essential building block for geometry

$$\begin{tikzcd} \mathcal{E} \arrow[d] \\ X \end{tikzcd} \iff \begin{tikzcd} E = \mathrm{Spec}(\mathrm{Sym}\,\mathcal{E}^\vee) \arrow[d] \\ X \end{tikzcd}$$

loc free sheaf — vector bundle. (v.b.)

The v.b. is itself a geometrical object.

Apart from that, we can construct many new objects/morphisms using v.b.:

E.g. 1. (linear series)

If $\mathcal{L} \in \mathrm{Pic}(X)$ is base-point-free, then it induces a map

$(m = h^0(\mathcal{L}))$

$$\begin{aligned} X &\longrightarrow \mathbb{P}^{m-1} = Gr(m-1, H^0(\mathcal{L})) \\ p &\longmapsto H^0(\mathcal{L} \otimes \mathcal{I}_p) \end{aligned}$$

Given $s \in H^0(\mathcal{L})$, gets

$$\begin{tikzcd} X \arrow[r] & Gr(m-1, H^0(\mathcal{L})) \\ X_s \arrow[u, phantom, "\cup"] \arrow[r, mapsto] & H_s \arrow[u, phantom, "\cup"] \end{tikzcd}$$

$\rightsquigarrow$ family of hypersurfaces; Lefschetz pencil

E.g. 2. (cyclic covers)

If $\mathcal{L}^{\otimes k} \cong \mathcal{O}_X$, then $\exists$ unramified cover $\pi: Y \to X$

s.t. $\begin{cases} \mathcal{O}_Y \cong \pi^* \mathcal{L} \\ \pi_* \mathcal{O}_Y \cong \bigoplus_{j=0}^{n-1} \mathcal{L}^{-j} \end{cases}$

$$\begin{tikzcd} Y \arrow[r, phantom, "\subset"] \arrow[dr, "\pi", "k:1"'] & L \arrow[d] \\ & X \end{tikzcd}$$

$Y = \{(x, s) \in L \mid s^k = 1\}$

(Also induce $\mathbb{P}^{r-1}$-bundles on X via similar way)

E.g. 3. (polarizations on AV)

When $X = A$ is an abelian variety, a l.b. $\mathcal{L} \in \mathrm{Pic}(A)$ induces

$$\begin{aligned} \phi_{\mathcal{L}} : A &\longrightarrow \hat{A} := \mathrm{Pic}^{\circ}(A) \\ p &\longmapsto t_p^* \mathcal{L} \otimes \mathcal{L}^{-1} \end{aligned}$$

E.g. 4 ...

Q: How are they related? How to understand these new objects/morphisms?

$$\begin{tikzcd} Y \arrow[d] & \\ X \arrow[r, two heads] \arrow[dr, hook] & \hat{A} \\ & \mathbb{P}^{m-1} \end{tikzcd}$$

What: Vect_X is not enough.

when we view Vect_X as a full subcategory of Coh_X, i.e, morphism matters

① Vect_X is not an abelian category $\rightsquigarrow$ Coh_X

E.g. For $D \subset X$ prime divisor, we have SES:

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \iota_{D,*}\mathcal{O}_D \longrightarrow 0$$

$$\mathcal{O}_X(-D) = \mathcal{I}_D$$

$\mathcal{I}_D$ & $\mathcal{O}_X$ are l.bs but $\iota_{D,*}\mathcal{O}_D$ is not.

(this SES is very useful, e.g., in the proof of RR for curves)

Rmk. To get more SES, we have to work in bigger categories.

e.g. we can not work on the de-Rham res

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega_X^1 \xrightarrow{d} \cdots \xrightarrow{d} \Omega_X^n \longrightarrow 0$$

in the category Coh_X, since d is not a morphism.

② $\mathrm{Coh}_X \rightsquigarrow D^b(\mathrm{Coh}_X)$ convenient for manipulation.

③ (semi) stable v.b.: "bounded family" for moduli spaces.

④ exceptional/spherical v.b.: special objects, (building block for v.b.)
helps to understand $D^b(\mathrm{Coh}_X)$

Which: task distributions.

$$\begin{tikzcd} & \mathcal{E} \arrow[dl, dash] \\ X & \end{tikzcd}$$

1. $\mathcal{E} = \mathcal{L} \rightsquigarrow \mathrm{Pic}(X)$

2. $X = \mathcal{C}$
   - $X = \mathbb{P}^1$ — $\mathcal{E} = \bigoplus_{a_i} \mathcal{O}(a_i) \xrightarrow{\mathrm{Coh}}$ Talk 6
   - $X = E$ — Talk 2 & Talk 3
   - $g(X) \geq 2$ — Talk 4
   - degree 0 — Talk 5

3. $\dim X \geq 2$
   - equiv v.b. on $Gr(r, n)$ — Talk 7
   - exceptional v.b. on $\mathbb{P}^2$ — Talk 9
   - surfaces, e.g. K3 — Talk 10 ...

Tool talk. — Talk 8

Chern characters.